

ZMEINOGORSK, Russian FederationWMO: **360380**Lat: **51.1550N**Lon: **82.1722E**Elev: **355**StdP: **97.14**Time Zone: **7.00 (E07)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.8	(c) -32.8	(d) -39.5	(e) 0.1	(f) -36.6	(g) -35.5	(h) 0.1	(i) -32.7	(j) 10.7	(k) -5.6	(l) 9.5	(m) -5.7	(n) 0.2	(o) 50	(p) 0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.6	(c) 30.8	(d) 18.7	(e) 29.0	(f) 18.1	(g) 27.4	(h) 17.3	(i) 20.4	(j) 27.7	(k) 19.4	(l) 26.5	(m) 18.6	(n) 25.2	(o) 2.5	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.5	23.5	16.9	12.6	22.2	16.0	11.8	21.5	60.6	27.9	57.0	26.6	54.2	25.3	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.4	-38.1	34.3	4.7	2.0	-41.5	35.7	-44.2	36.9	-46.8	38.0	-50.2	39.5		
			-38.2	22.0	4.4	2.0	-41.4	23.5	-44.0	24.7	-46.4	25.8	-49.6	27.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-14.9	-12.8	-5.2	5.3	12.2	17.8	19.3	17.2	11.3	4.7	-4.7	-11.2	(6)
(7)		DBStd	13.53	8.59	8.34	6.83	5.82	4.87	3.82	3.09	3.60	4.64	5.31	8.04	7.70	(7)
(8)		HDD10.0	3391	771	637	471	159	34	2	0	1	38	177	442	659	(8)
(9)		HDD18.3	5606	1029	871	729	393	196	54	26	63	214	423	691	917	(9)
(10)		CDD10.0	959	0	0	0	17	103	235	288	224	78	13	1	0	(10)
(11)		CDD18.3	132	0	0	0	0	7	38	55	29	4	0	0	0	(11)
(12)		CDH23.3	1746	0	0	0	9	151	443	573	455	113	2	0	0	(12)
(13)		CDH26.7	488	0	0	0	1	37	113	164	141	32	0	0	0	(13)
(14)	Wind	WSAvg	2.4	2.6	2.5	2.5	2.6	2.6	2.1	1.8	1.8	1.9	2.6	3.0	3.1	(14)
(15)	Precipitation	PrecAvg	664	44	41	39	48	64	56	73	66	53	71	66	54	(15)
(16)		PrecMax	1010	165	111	121	124	147	170	163	176	156	194	133	118	(16)
(17)		PrecMin	498	0	3	6	5	2	2	12	12	3	24	6	6	(17)
(18)		PrecStd	114	32	22	26	26	37	37	41	43	31	38	29	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	4.4	12.5	24.6	30.9	31.7	33.4	33.6	30.4	23.0	14.5	4.5	(19)
MCWB			1.4	0.8	5.7	12.4	16.2	18.4	20.8	19.8	16.2	11.4	7.1	0.5	(20)	
2%		DB	1.1	2.2	8.4	21.1	27.2	29.5	30.3	30.3	26.6	19.1	10.4	1.9	(21)	
		MCWB	-1.0	-0.5	3.6	10.8	14.6	18.1	19.7	18.5	14.5	9.9	5.3	-0.5	(22)	
5%		DB	-1.3	0.3	5.9	18.3	24.9	27.8	28.4	28.1	23.5	16.1	7.8	0.4	(23)	
		MCWB	-2.9	-1.6	2.1	9.6	13.7	17.5	19.0	17.5	13.7	8.7	4.0	-1.5	(24)	
10%		DB	-3.4	-1.4	4.1	15.4	22.4	26.0	26.7	25.8	20.7	13.3	5.0	-1.4	(25)	
		MCWB	-4.8	-3.2	1.3	8.3	12.8	17.3	18.3	16.8	12.5	7.6	2.2	-2.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	1.5	6.2	13.2	17.6	20.7	22.9	21.2	17.6	12.2	8.2	1.1	(27)
MCDWB			4.3	3.5	11.6	22.7	27.1	27.6	30.2	30.3	27.2	20.3	12.4	2.9	(28)	
2%		WB	-0.8	-0.1	4.1	11.6	16.0	19.5	21.1	19.5	15.4	10.5	5.7	-0.1	(29)	
		MCDWB	0.9	1.6	7.7	19.1	24.3	27.0	27.9	27.6	24.3	17.0	9.9	1.7	(30)	
5%		WB	-2.8	-1.6	2.7	10.2	14.8	18.6	19.9	18.4	14.2	9.3	4.0	-1.4	(31)	
		MCDWB	-1.4	0.4	5.3	16.9	22.0	25.8	26.5	26.0	22.1	15.1	7.4	0.3	(32)	
10%		WB	-4.8	-3.2	1.4	8.8	13.7	17.8	19.0	17.4	13.0	7.9	2.3	-2.9	(33)	
		MCDWB	-3.5	-1.4	3.9	14.9	20.7	24.3	25.2	24.3	19.7	13.0	5.1	-1.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.9	11.6	10.7	11.7	13.4	12.2	11.6	13.5	13.2	10.4	8.4	9.4	(35)
MCDBR			9.6	10.9	12.0	17.7	18.5	16.0	15.4	17.7	19.2	16.8	10.3	9.3	(36)	
MCWBR			8.2	8.8	8.1	9.3	8.1	6.7	6.5	7.4	8.9	9.1	6.8	7.8	(37)	
5% WB		MCDWB	8.9	10.5	10.1	15.8	14.4	13.0	12.8	15.1	17.2	15.0	9.5	9.0	(38)	
		MCWBR	7.8	8.7	7.1	8.8	7.0	6.2	6.1	7.2	8.4	8.5	6.5	7.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.336	0.386	0.420	0.416	0.411	0.433	0.406	0.367	0.359	0.340	0.305	(40)	
taud		2.019	1.948	1.905	2.080	2.165	2.257	2.200	2.255	2.311	2.214	2.051	1.972	(41)		
Ebn at Noon		671	748	783	814	839	846	818	822	815	730	610	580	(42)		
Edh at Noon		85	121	154	146	141	130	135	122	103	93	83	76	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.37	2.48	3.87	4.61	5.83	6.30	5.74	5.19	3.81	2.06	1.15	0.96	(44)	
(45)		RadStd	0.10	0.13	0.36	0.39	0.47	0.53	0.42	0.31	0.33	0.27	0.12	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	-0.94	-1.15	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	-0.46	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page